AIM

PT.130 Performance Test Report

<Company Long Name>

<Subject>

Author:	<Author>
Creation Date:	May 21, 1999
Last Updated:	August 15, 2025
Document Ref:	<Document Reference Number>
Version:	DRAFT 1A

Approvals:

<Approver 1> ______________________________

<Approver 2> ______________________________

ORACLE

Copy Number _____

Document Control

Change Record

Date	Author	Version	Change Reference
21-May-99	<Author>	Draft 1a	No Previous Document

Reviewers

Name	Position

Distribution

Copy No.	Name	Location
1	Library Master	Project Library
2		Project Manager
3		
4		

Note To Holders:

If you receive an electronic copy of this document and print it out, please write your name on the equivalent of the cover page, for document control purposes.

If you receive a hard copy of this document, please write your name on the front cover, for document control purposes.

Contents

Executive Summary

Oracle Applications™ is a suite of advanced, open systems, application products that <Company Short Name> has implemented to manage parts of its enterprise business information. These products use the Oracle Relational DBMS technology to store data and their design provides comprehensive and complex functionality.

Execution of a performance test project took place between the dates of <Test Project Start Date> and <Test Project End Date>. Performance testing assists the Oracle Applications implemented at <Company Short Name> to perform the business processing within an acceptable time frame. Design of this test took into consideration the expected performance of the applications when processing in a business environment similar to <Company Short Name>'s live production systems. This report describes the method and results of the Performance Test performed at <Company Short Name>.

Summary of Test Results

The main results obtained from the test were:

-
-
-

Conclusions and Recommendations

The conclusions drawn from the test results are:

-
-
-

In the light of these results, the leader of the performance test team makes following recommendations.

1.
2.
3.

Project Scope, Objectives, and Approach

Scope

The Performance Test conducted for <Company Long Name> included the following Oracle Applications:

- <Application 1>
- <Application 2>
-

The business processes represented in the test were:

- <Process 1>
- <Process 2>
-

Objectives

The purpose of the Performance Test was to …

Target Performance Metrics

The target metrics for the performance test were as follows:

Performance Measurement	Type of Measurement	Target Metric	Current Metric	Description

Approach

In order to prepare a detailed test plan and testing approach, the performance test team has prepared an analysis that breaks out processing loads and identifies tangible measurements to gauge system performance.

Selection of transaction flows included interviews with <Company Short Name> business analysts and IS staff. The intention was to model potential peak load periods, thereby ensuring acceptable performance during critical processing time frames. Each of these special peak load processing periods, termed performance test scenarios, will have its own characteristic relative mix of online and batch usage.

The precise specification of mixes for online and batch transactions exist as particular transaction models for each scenario. Each transaction model has very precise numbers of simulated test users and sessions, and each test user also has a precisely defined rate at which to perform transactions.

The analysis also provides information about the size of the test servers necessary to mimic the production environment. Accurate server sizing relative to the live production servers is essential if the performance results are to be a fair reflection of the system response in the production environment.

Performance Test Strategy

Methodology

The method chosen for this subproject adheres to the *Performance Testing* process within Oracle's Application Implementation Method.

The AIM *Performance Testing* process defines the following tasks and deliverables.

ID	AIM Phase	Task Name	Deliverable	Description
PT.010	Definition	Define Performance Testing Strategy	Performance Testing Strategy	
PT.020	Operations Analysis	Identify Performance Test Scenarios	Performance Test Scenarios	
PT.030	Operations Analysis	Identify Performance Test Transaction Models	Performance Test Transaction Models	
PT.040	Solution Design	Create Performance Test Scripts	Performance Test Scripts	
PT.050	Solution Design	Design Performance Test Transaction Programs	Performance Test Transaction Program Designs	
PT.060	Solution Design	Design Performance Test Data	Performance Test Data Design	
PT.070	Solution Design	Design Test Database Load Programs	Performance Test Database Load Program Designs	
PT.080	Build	Create Performance Test Transaction Programs	Performance Test Transaction Programs	
PT.090	Build	Create Test Database Load Programs	Performance Test Database Load Programs	
PT.100	Build	Construct Performance Test Database	Performance Test Database	
PT.110	Build	Prepare Performance Test Environment	Performance Test Environment	
PT.120	Build	Execute Performance Test	Performance Test Results	
PT.130	Build	Create Performance Test Report	Performance Test Report	

Automated Testing Tools

<Describe the selection and use of testing tools here>

Test Execution

<Describe test execution methods here>

Test Models

Test Scenarios

Each test scenario consists of a number of active business functions. The business functions assumed to be active in the test scenario are the Test Business Functions. Business Function Roles perform these Test Business Functions.

Test Scenario 1 - <Performance Test Scenario Name>

This scenario corresponds to a system processing situation…<enter description of situation here>

The composition of this scenario was:

Test Business Function	Function Type	Business Function Role	Current # of Users	# of Users as of <Business Metrics Date>
Order Entry	Online	Order Administrator	53	70

with the following processing metrics:

Test Business Function	Business Metric	# of Users	Rate	Comments
Order Entry	1400 new orders in 5 hours	70	1 order per user per 15 min	Assuming 10% growth in sales orders

Transaction Models

Each test scenario consists of a series of transactions corresponding to the business functions and user roles identified in the test scenarios.

<Enter the detailed transaction model information here if appropriate>

Test Scripts

We defined test user roles in order to process the transactions of each specified scenario. Each test user role associates to a simulated number of users in a test cycle. Thus, you may think of a test user role as a conceptual application user process flow. A real system user may map onto one or more test user roles.

To implement the process flow for each test user role, transactions associated with the flow decompose into one or more test scripts.

Test Database

Database Population Strategy

Data Volumes

In deciding the volume of data to use in the test database it is necessary to take into account purge/archive cycles for the different application data. Since these cycles may differ, the historical volume of data may vary between different data entity tables at any given point in time. The figures below represent the actual volumes of data used in the test database. These figures reflect the *worst* possible processing situation (highest volume in the database) with the purge/archive cycles that <Company Short Name> has specified.

Business Object	Transaction Rate (Objects/month)	Historical Volume (months)	Total Data Volume (Objects)

System Architecture and Configuration

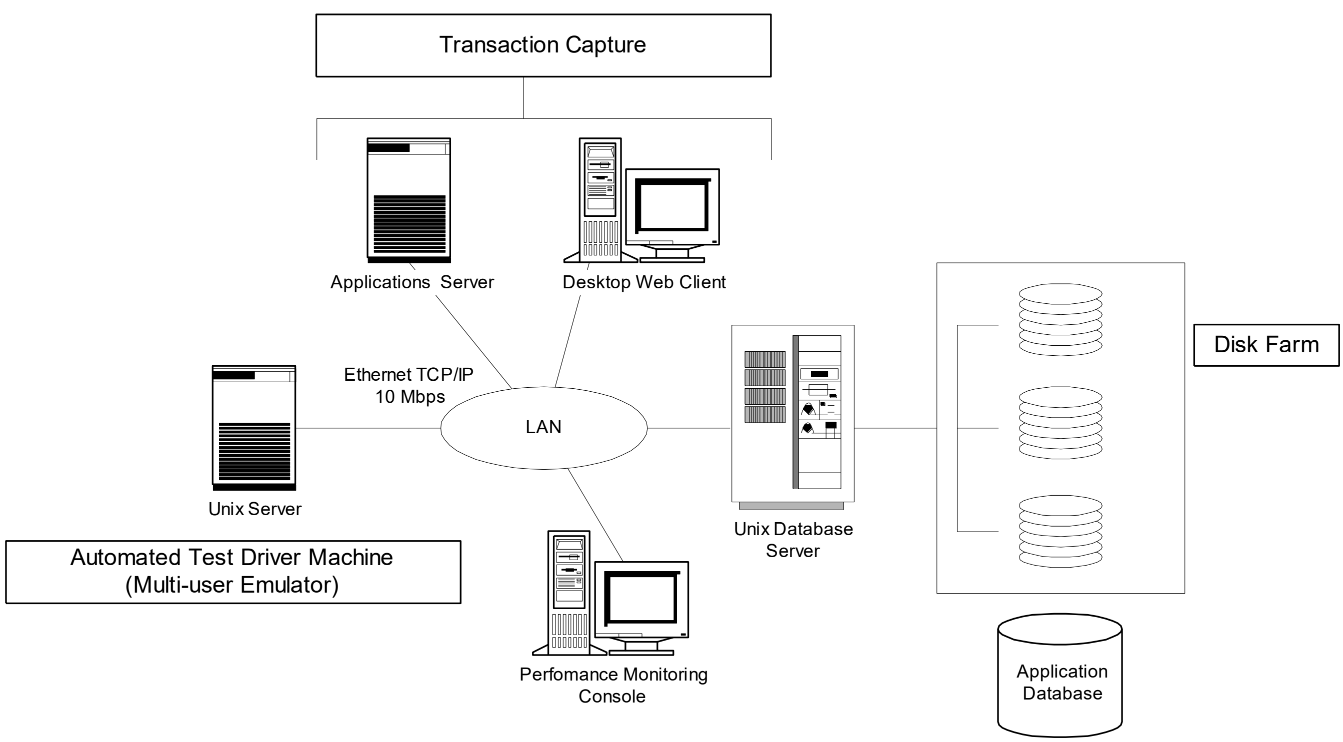
This section describes the system architecture and configurations implemented for the performance test. It includes detailed specifications for the technical infrastructure of the test, that is, hardware, operating system and network components, as well as the application configuration.

Technical Architecture

Specifications for the technical architectures included in testing appear below.

Configuration 1 - Central Host

Hardware, O/S and Network Configuration



Database Configuration

Overview

ORACLE ID Configuration

Instance	ORACLE ID	Responsibility
PERFTEST	APPLSYS	System Administrator
	GL	GL Super User
		GL Clerk

Init.ora Parameters

Instance	Parameter Name	Value
PERFTEST	DB_BLOCK_BUFFERS	
	SHARED_POOL_SIZE	

Database Object Distribution

Tables

Instance	Table Name	Tablespace	Table Size	Disk Drive

Indexes

Instance	Index Name	Tablespace	Index Size	Disk Drive

Application Configuration

Key Set-Up Parameters

Application Object Library

Standard Options Used

Custom Options Used

Assumptions

Business Model

<Enter assumptions about the future business model here>

Application Functionality and Configuration

<Enter assumptions about the use application functionality and configuration here>

Example: Test Scenarios 1 and 2 concern themselves, in part, with key components of the overall order-to-fulfillment process. In defining these test processes, the assumption will be made that the BOMs will be managed using an PTO model in Oracle and will not use the ATO functionality

Technical Architecture

<Enter assumptions about the future technical architecture here>

Risks

Performance Test models used are artificial constructs since it is impossible to exactly simulate all the complex interactions and processing that takes place in a production ERP system. There are inherent risks in the interpretation of the results from a test that only includes a small subset of the processing that actually occurs in a particular situation. The limited range of transactions and business processes included in the scenarios may not sample all processes currently recognized to create a large performance overhead. Nor can the test include processes subsequently found to be of sufficient concern to warrant their inclusion in such a test.

Resource and time limitations mean inclusion of only a small number of scenarios in the testing. These scenarios provide coverage to at least some of the concerns of the finance, supply and demand teams. This does not represent a complete list of situations in which the combined processing of the separate users and batch processes create an usual load on the hardware. The project team recognizes that there other scenarios worthy of test, given sufficient resource and time.

Test Results

Summary

<Summarize the test results>

Performance Measurements

This section lists the test performance measurements taken during each scenario. The measurements are of two general types; the first involve measuring response times for particular events that occur during the transaction processing of a scenario. An example is the time taken for a navigation event, saving changes to the database or displaying a list of values. The other type involves making measurements about gross system performance such as CPU usage or network load. When a measurement does not associate with a particular system business event, the transaction column is blank.

Scenario	Transaction	Meas. Code	Measurement
1	Display List of Customers in Enter Customer Form	A-LOV-01	Time for list to appear after keystroke to display list
		A-CPU-01	CPU % Usage

Issues Identified and Resolution

Application Processing

Patches Applied

No.	Type	Patch Reference	Bug Number	Impact to Performance Test	Date Executed	Resp.

Conclusions and Recommendations

Conclusions

<Enter the conclusions drawn from the test results here>

Recommendations

<Enter the recommendations arising from the test here>

Appendix A: Glossary

Benchmark Test

A test that measures the limiting performance metrics for a selected configuration and prescribed set of application test functions, as one or more ***test parameters*** will vary towards some limiting value. Is usually concerned with testing a small number of special transactions in a highly tuned environment, rather than simulating the complex business processing of an enterprise in live production. In the case of a published benchmark, an impartial external agency (such as TPC benchmarks) prescribes the test transactions.

Performance Test

A test which measures performance metrics for a hardware/software configuration that is processing a set of application test functions subject to a single set of ***test parameters***. Usually the design of the test is to mimic a particular production processing environment. It differs from a ***benchmark*** in that a ***benchmark*** is less likely to consider simulating a production environment.

Performance Test Scenario

A point-in-time sample of the processing environment to be tested. Each scenario will include, users working in multiple applications, with on-line and/or batch processing being performed.

Test Cycle

An instance of a test ***transaction model*** with a specific set of ***test parameters*** or a specific hardware configuration. For example, by varying the number of simulated users in multiple cycles, it is possible to determine the capacity of the configuration under test.

Test Parameters

Test parameters are the set of test variables that may alter during a series of ***test cycles***, possibly for ***benchmarking*** purposes. Test parameters can change without altering the hardware/software configuration or the mix of transactions that comprise the overall application test functions. Examples are the total number of users being simulated in the test or the number of CPU slots utilized in a server.

Test Scenario

A description of the processing situations (point-in-time processing snapshots) in the future production system where there will be critical performance requirements. These performance ***test scenarios*** form the basis of models of the system transactions that make up the test.

Test Script

A program created to execute all or part of a specific ***Test User Role***. If the ***test user role*** models a real application process flow in a ***test scenario***, a component test script will cause the execution of transactions for a realistic process flow.

Test User Role

A set of transactions associated with a defined group of application users in the ***Performance Test***. A ***test user role*** may include a set of transactions that attempt to model a "real" application user process active during a ***test scenario***. In some cases, a test user role facilitates segregation of the total active process flows, using a heuristic transaction grouping scheme. Transaction(s) process for each ***test user role*** through the execution of one or more unique ***test scripts***.

Transaction Model

A specific mix of transactions for a particular ***test scenario***. Each ***test scenario*** is supported by one or more ***transaction models***.

Open and Closed Issues for this Deliverable

Open Issues

ID	Issue	Resolution	Responsibility	Target Date	Impact Date

Closed Issues

ID	Issue	Resolution	Responsibility	Target Date	Impact Date